



U.S. General Services
Administration



Agentic AI Book Club - Week 11

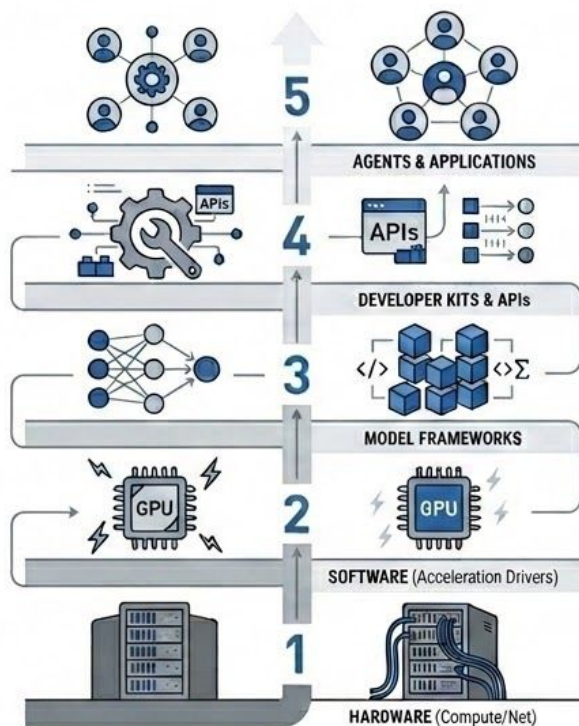
Reviewing Chapter 2.7, 2.8 & 2.9

AI Community of Practice - 2026

Housekeeping

- **Agentic AI is an early and rapidly evolving knowledge domain**
 - Google form to collect take-home test will be provided next week
 - No meeting until July (we'll send out communications)
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- A decorative graphic in the bottom right corner consisting of several overlapping, semi-transparent blue geometric shapes, including rectangles and triangles, arranged in a layered, ascending pattern.

NVIDIA Production Stack -- Five Layers



APPLICATION

NIM, NeVA & DePlot (Vision)

ORCHESTRATION

(Combined with APPLICATION for simplicity in graphic flow)

MODEL

LLMs, **Multimodal** Embeddings)

STORAGE

Milvus (Accelerated Search)

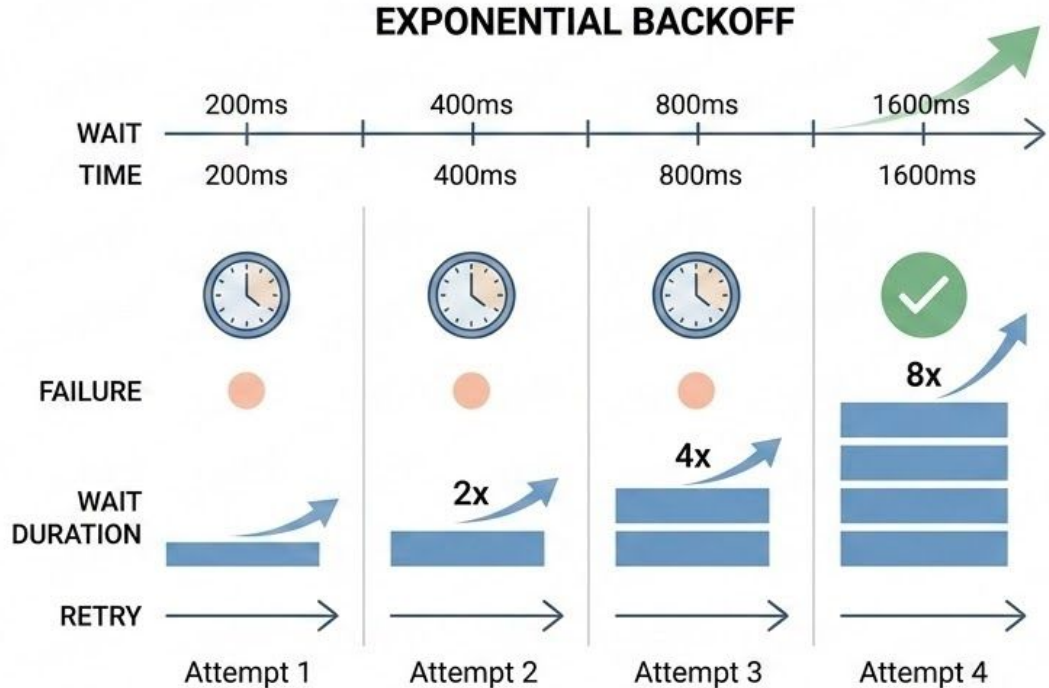
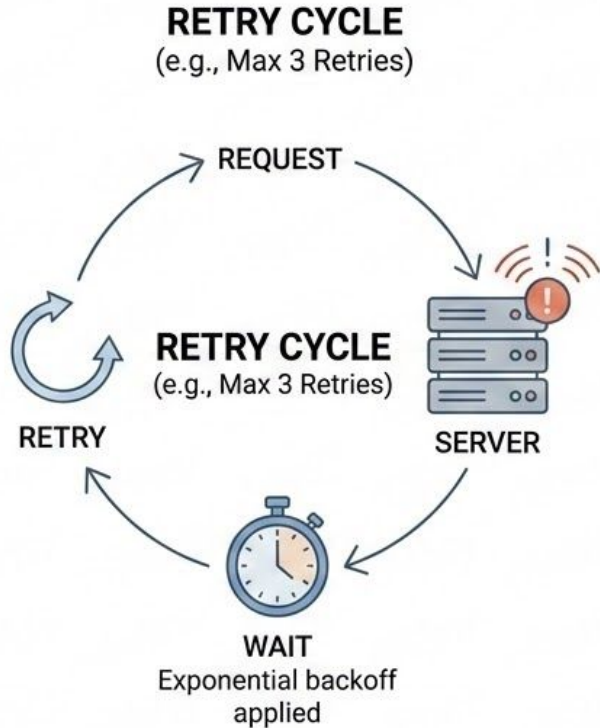
INFRASTRUCTURE

GPUs

Key Takeaways from Chapter 2.7

- Three architectures: unified, ground-to-text, separate stores
 - Match vision models to content complexity
 - Audio via Whisper adds a critical knowledge dimension
 - NVIDIA stack turns prototypes into production systems
 - Next: Chapter 2.8 on error handling and resilience
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Retry Logic and Exponential Backoff



Fallback Strategies and Model Routing

- Route to alternative providers when primary fails
 - Model routing: GPT-4o to Claude to cached results
 - Log fallback events for capacity planning
 - Human-in-the-loop escalation for high-stakes decisions
 - Combine retries first, then fallback if retries exhaust
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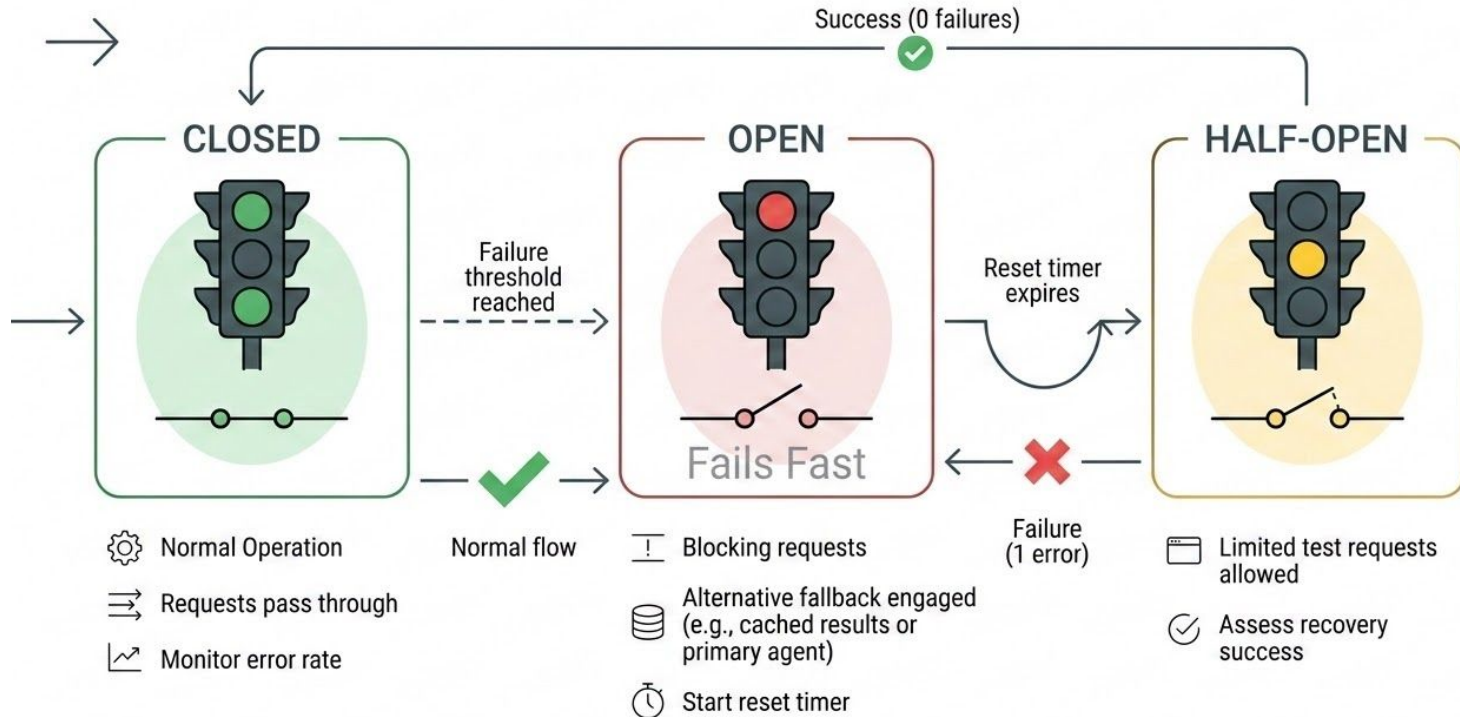
Fallback Pitfalls to Watch For

- Shared failure domains defeat fallback purpose
 - Reactive checking adds 2-5 seconds of latency overhead
 - Fallback providers differ in cost and output quality
 - Price profile differences
 - Silent quality changes erode user trust over time
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Graceful Degradation Patterns

- Distinguish core vs. auxiliary capabilities
 - Core: minimum viable functionality users need
 - Auxiliary: enhancements that are nice but not essential
 - Disable auxiliaries first, preserve core functions
 - Research agent returns links even when summaries fail
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Circuit Breakers -- Three States

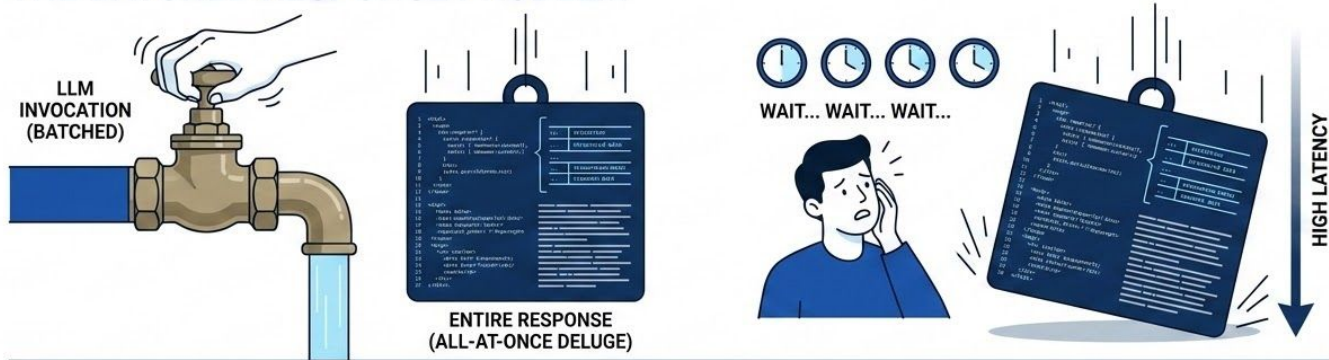


Framework Support -- LangChain and LangGraph

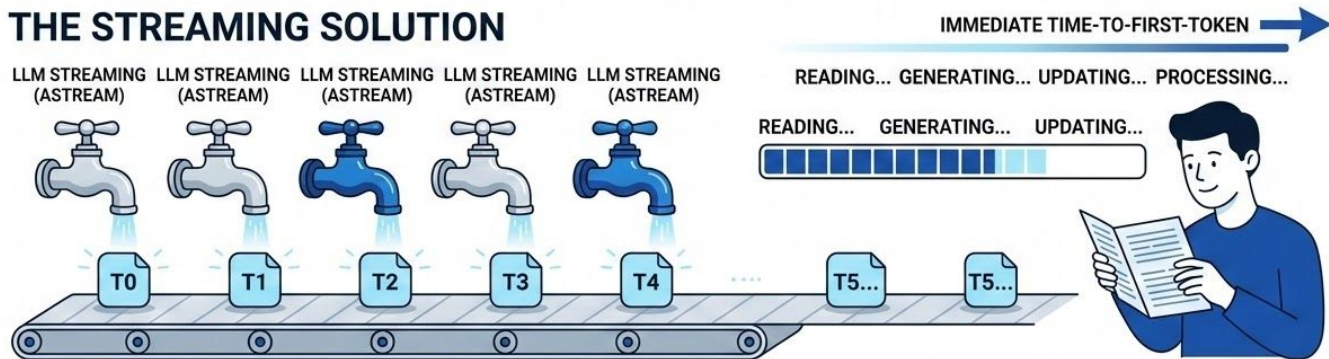
- LangChain AgentExecutor has built-in retry configuration
 - `handle_parsing_errors` recovers from malformed outputs
 - `max_execution_time` prevents runaway agent resource drain
 - LangGraph conditional edges model explicit recovery paths
 - Wrap framework agents in circuit breakers for full resilience
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The Batched Response Problem & Streaming

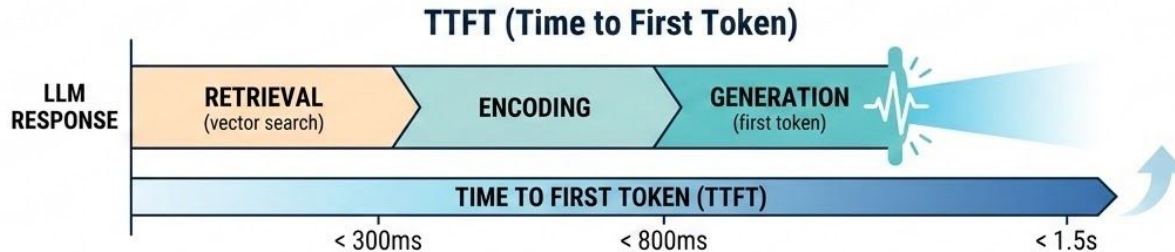
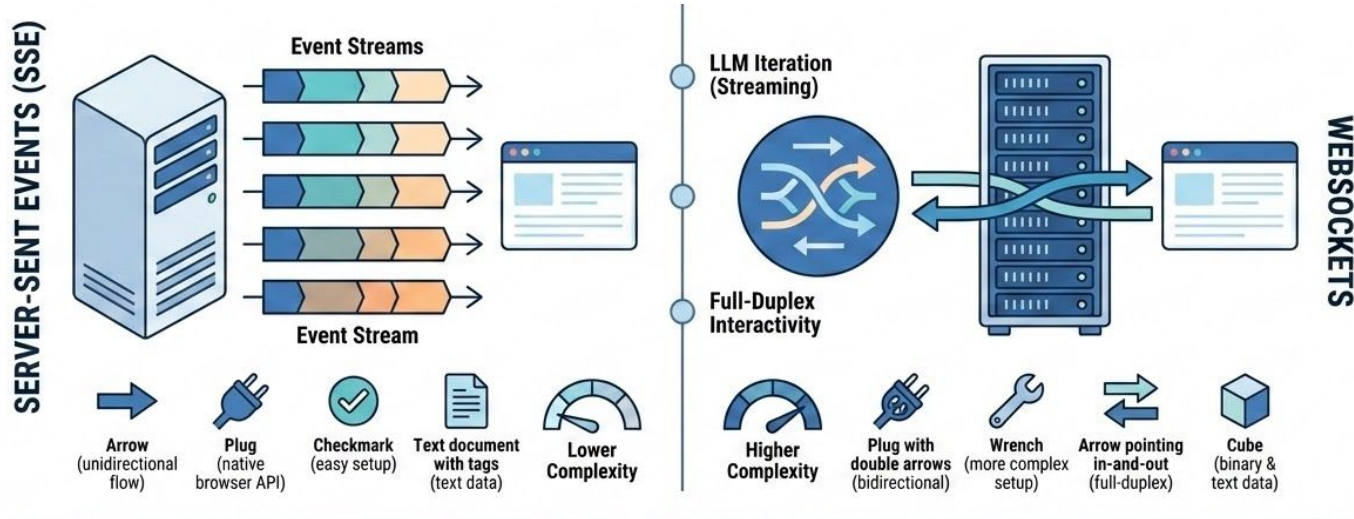
THE BATCHED RESPONSE PROBLEM



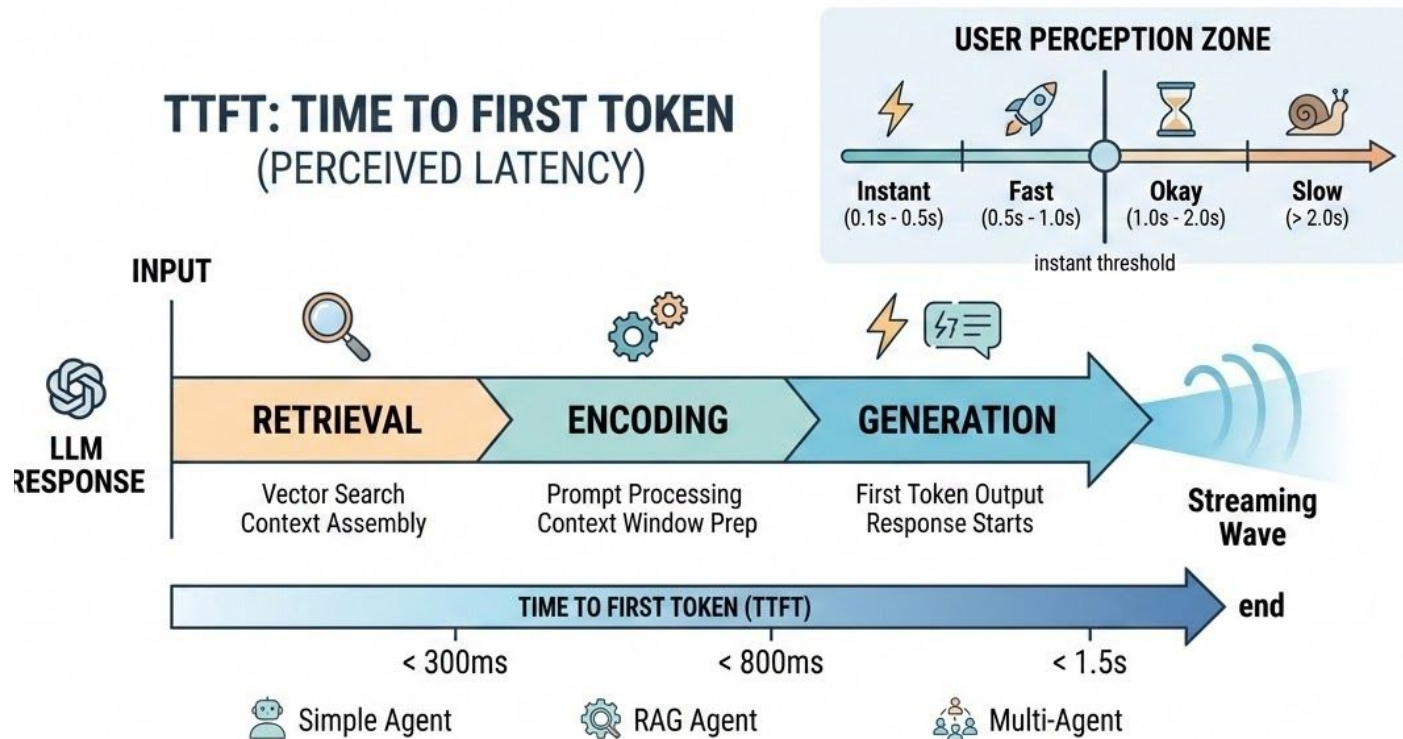
THE STREAMING SOLUTION



SSE vs. WebSockets



Time to First Token -- The Metric That Matters



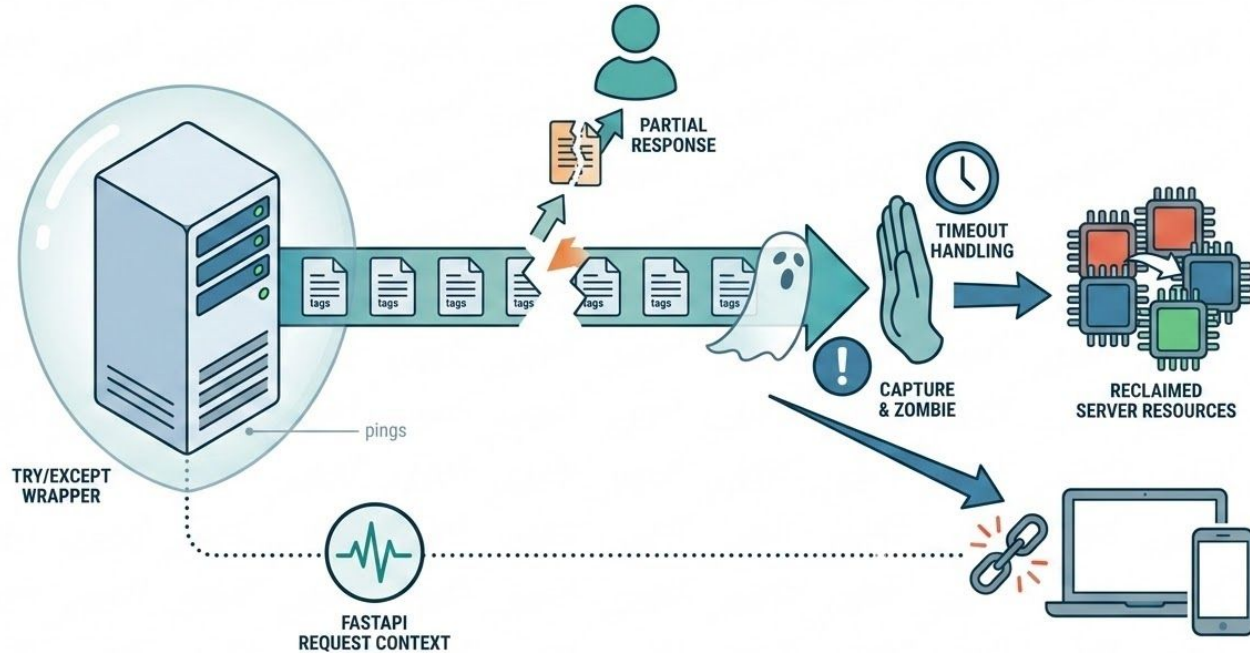
Optimizing TTFT Systematically

- Parallel hybrid retrieval cuts search time by 50%
 - Semantic caching serves repeated questions in under 200ms
 - Prompt compression reduces encoding latency by 60-70%
 - Faster models for initial response, powerful models for follow-up
 - NVIDIA NIM on-premises eliminates cloud round-trip latency
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TTFT Targets and Streaming Metrics

- Simple Q&A agents: 300-500ms TTFT target
 - RAG agents with vector search: 800ms-1.2s target
 - Multi-agent systems: 1.5-2s target
 - Above 2 seconds enters "slow" perception territory
 - Log retrieval, encoding, queue, and network times separately
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Production Robustness



LangServe -- Production Streaming Infrastructure

- LangServe auto-generates `/stream`` and `/invoke`` endpoints
 - Single `add_routes()` call exposes any chain with streaming
 - `/stream_log`` shows intermediate agent reasoning steps
 - Built-in retry logic, fallback models, and error handling
 - Standardizes API patterns across all deployed agents
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See you in Section 2